Quiz 7

1. Consider the following solutions to a differential equation set by a differential operator D:

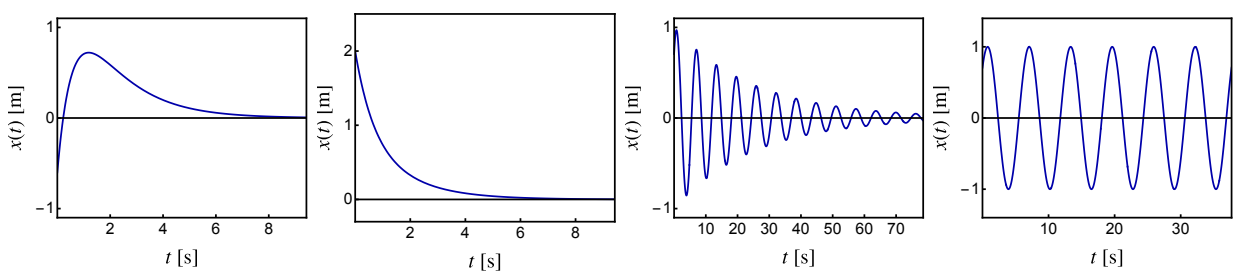
$$Dx_p(t) = f(t) \tag{1}$$
$$Dx_h(t) = 0 \tag{2}$$

Which of the following are solutions to Eq. (1)?

(a) $x_p(t)$
(b) $x_h(t)$
(c) $x_p(t) + x_h(t)$

2. Label each of the plots in Fig. 1 as one of the following:

(i) Overdamped harmonic oscillator
(ii) Underdamped harmonic oscillator
(iii) Undamped harmonic oscillator



(a) (i) (b) (i) Figure 1 (c) (ii) (d) (iii)

3. Consider a driven oscillator with the following amplitude:

$$A^2 = \frac{f_0^2}{\left(\omega_0^2 - \omega^2\right)^2 + 4\beta^2\omega^2}$$

Assuming ω_0 is fixed, what value of ω gives a resonance?

$$\frac{\partial}{\partial\omega}(\text{denominator}) = \frac{\partial}{\partial\omega}\left[(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2\right] = 2(\omega_0^2-\omega^2)(-2\omega) + 8\beta^2\omega$$

$$0 = -4\omega(\omega_0^2-\omega^2) + 8\beta^2\omega$$

$$0 = -\omega_0^2 + \omega^2 + 2\beta^2$$

$$\Rightarrow \boxed{\omega^2 = \omega_0^2 - 2\beta^2} \quad \text{or} \quad \omega = \sqrt{\omega_0^2 - 2\beta^2}$$